

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EST200-ANSWER SCHEME
Course Name: Design and Engineering

Max. Marks: 100

Duration:
3 Hours

PART A

Answer all questions. Each question carries 3 marks

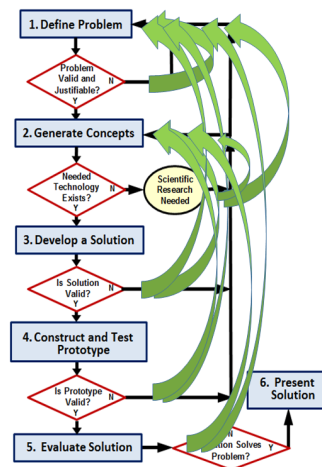
Marks

- 1 What are the basic vocabularies in engineering design? (3)
Design attributes, Design Function and Design Constraints, Ergonomics, Value Design, Aesthetics, Modular, Biomimetic, Eco friendly, Recyclable, Reverse Engineering etc
- 2 How to identify the customer requirements of design? (3)

1. By taking customer feedback through survey, mail, direct questionnaire
2. By observing the change trends in customer preference
3. By seeing sales volume
- 3 Describe the iterative process involved in design thinking approach. (3)

Design Process

- Iterative



- 4 Describe the importance of the empathize phase in design thinking. (3)
Become one among the customer or putting ourselves in customer role and see how will I choose a product of my company as a customer

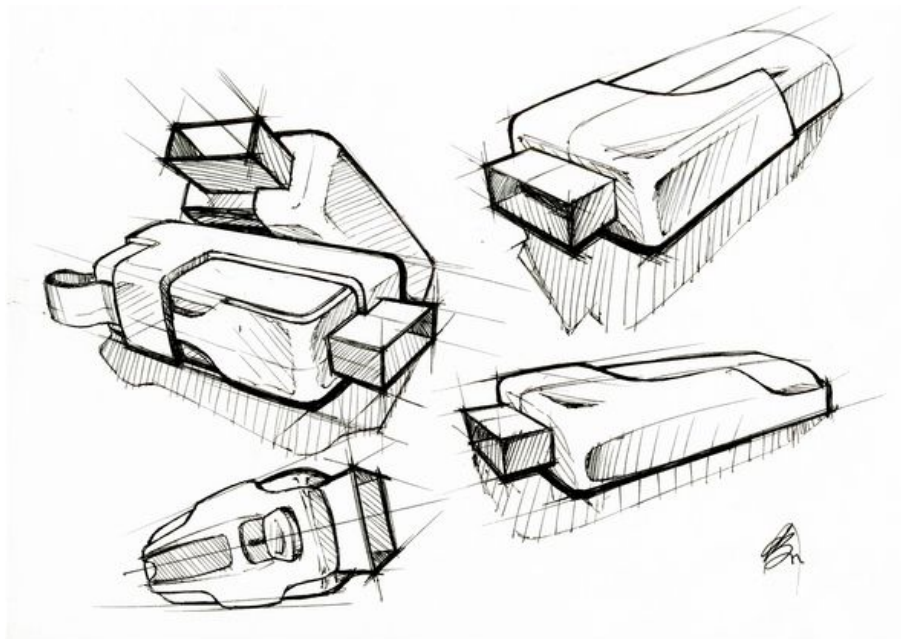
Empathy phase helps to:

- Appreciate people's emotional and physical needs.
- Gain insight into the way people see, understand and interact with the world around them.
- Realize how lives are impacted within the contexts being investigated.
- Find out what people mean rather than just what they say — empathic research is inherently subjective and is concerned with motivations and thoughts, rather than facts.

- 5 How can a design be communicated through engineering sketches and drawings? (3)

2d/3d representations using CAD

- Any product ideas can be effectively developed by appropriately sketching. For example a design of a USB flash drive is well elucidated and understood once the design idea is sketched, as shown below.



- 6 Explain the role of Prototyping in evaluating a Design

Evaluating a prototype allows the production costs to be assessed and finalized. Component failure is often identified during the testing process. This may mean a component is redesigned and not the entire product

7 Explain the importance of project-based learning in design engineering. (3)

- Collaboration: Relationships formed during collaboration is a huge part of PBL. Not only do students learn how to work better in groups—providing their own input, listening to others, and resolving conflicts when they arise—they build positive relationships with teachers, which reinforces how great learning is. Students also form relationships with community members when working on projects, gaining insight for careers and beyond.
- Problem Solving: Students learn how to solve problems that are important to them, including real community issues, more effectively—even learning from failure and possibly starting over.
- Creativity: Students apply creative thinking skills to innovate new product designs and possibilities for projects.
- In-Depth Understanding: Students build on their research skills and deepen their learning of applied content beyond facts or memorization.
- Self-Confidence: Students find their voice and learn to take pride in their work, boosting their agency and purpose.
- Critical Thinking: Students learn to look at problems with a critical thinking lens, asking questions and coming up with possible solutions for their project.
- Perseverance: When working on a project, students learn to manage obstacles more effectively, often learning from failure and making adjustments until they're satisfied with their work.
- Project Management: Students learn how to manage projects and assignments more efficiently.
- Curiosity: Students get to explore their curiosities, ask questions and form a new love for learning.
- Empowerment: Students take ownership over their projects, reflecting on and celebrating their progress and accomplishments.

8 Discuss the role of life cycle design approach in design decisions. (3)

A Life Cycle Design is a process of design and engineering work for the entire project and plant lifecycle from the initiation through project construction and production until disposal of the plant facility. The Life Cycle Design is integrated technology, people, process and business into the EPC project including operations and maintenance that identifies opportunities to increase the reliability and safety.

9 What are the factors to be considered for a sustainable design? (3)

Environmental Issues and design
Global climatic change
Environmental Life cycle assessment

10 What are design rights, and how can an engineer put it into practice? (3)

- 1) Exclusive right to apply design to the article in the class in which design is registered.
- 2) Better protection and can sue for piracy of design.
- 3) Licensing of design as legal property for consideration or royalty

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11 Find the customer requirements for designing a website for an educational institution. Show how the design objectives were finalized considering the design constraints. Sketch a layout of the website showing dropdown menus. (14)

Five essential components are

1. Thoughtful, Simple Navigation

It is essential that colleges provide website visitors with a straightforward and intuitive way to navigate a site and find the content they are looking for. Addressing multiple audiences' unique pain points and content needs can be a challenge for many institutions.

For example:

- Current Students may need know how to set up their portal account to access their class information
- Prospective Students may want information on student housing or scholarships
- Parents may want to learn more about campus safety

To address all of these needs, college websites need a clearly defined navigation structure so that key audiences can quickly gain access to the information that is most important to them.

Let's take a moment to explore a few of the most common ways a college can structure their navigation:

- Audience segmentation
- Main topic organization
- Hybrid approach

2. Robust Search Functionality

Often users visit a website to find information on a very specific topic. Website visitors expect to see a search box to assist them in finding content quickly. Universities need to have robust search functionality so that users can swiftly and easily access the content that they are looking for. Keep in mind that visitors expect certain key elements of a site—like a search bar or portal login—to be located in the same general area. Be sure to place your search bar strategically, and be consistent in this placement across all website pages.

3. Mobile Responsiveness

Today's university websites must be mobile responsive! Parents, faculty and students alike increasingly rely on their smartphones and tablets to access content on the web. If your college campus – like most – serves Millennial students in the 18-34 year old range don't delay in ensuring that your site is mobile responsive today!

When verifying the responsiveness of a college website make sure that the menu navigation is presented in a clear format for mobile. Often, elaborate drop down menus do not display clearly via mobile devices and provide a poor user experience to mobile visitors. Identify the components of your site that don't translate well on mobile devices and make sure you are still providing users with information they need. (Mambo can help sort this out with you.)

4. Include Calls To Action (CTAs)

Lead your visitors to engage with you. Are you using buttons and/or hyperlinks to drive your audience to take action? Have you identified what your calls to actions (CTAs) are and what you want your user to do on your pages? Do you want them to schedule a campus visit, request more information or find out more about your programs?

Once you identify what your CTA should say, you need to make sure it stands out on the page. Using contrasting colors and brief text helps attract a visitor's attention. Be sure to lead with specific CTAs throughout your website.

example sketches

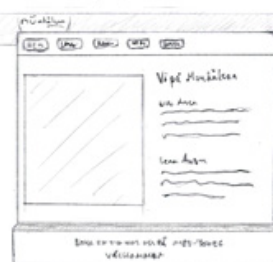
home page



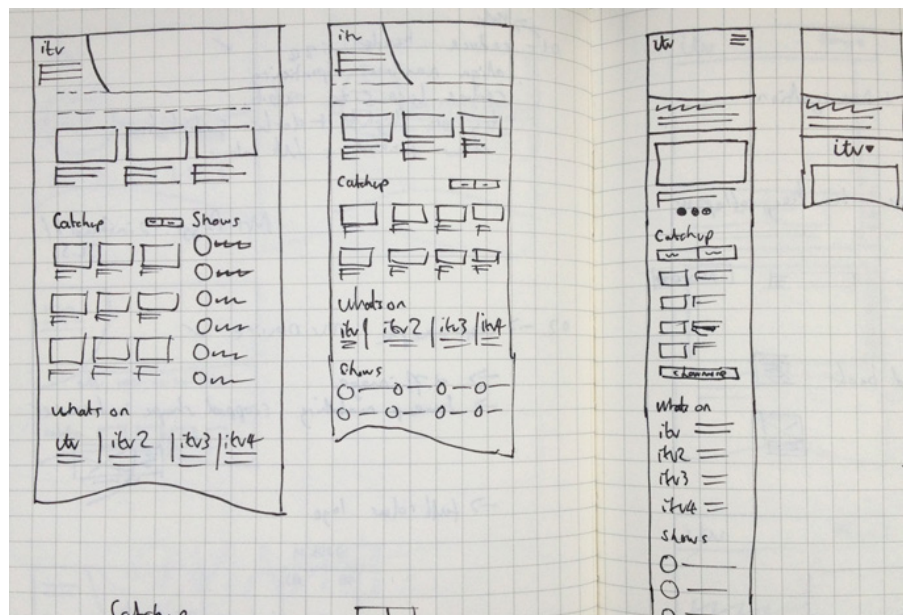
content page with submenu



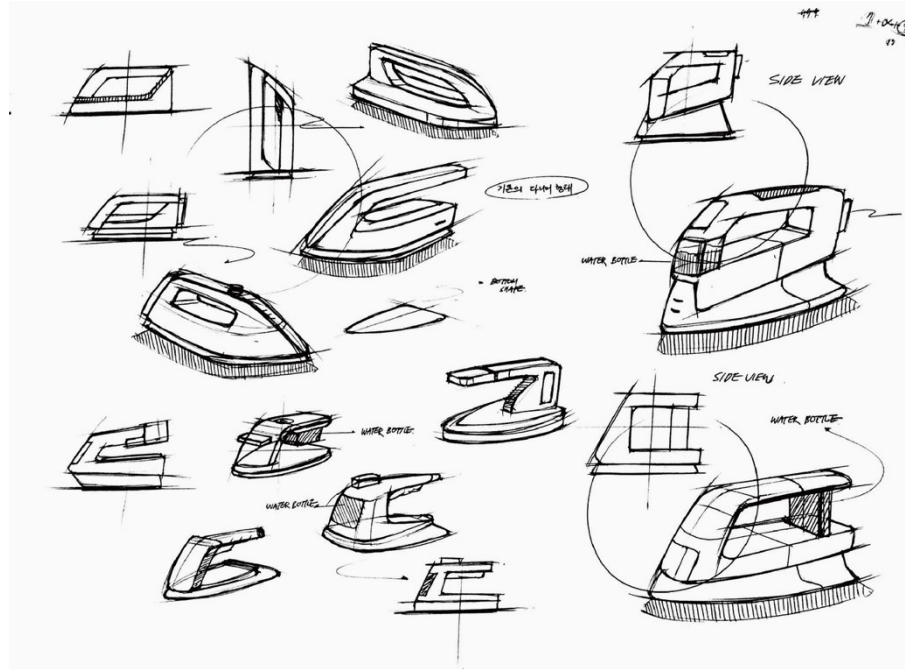
content page



responsive mobile



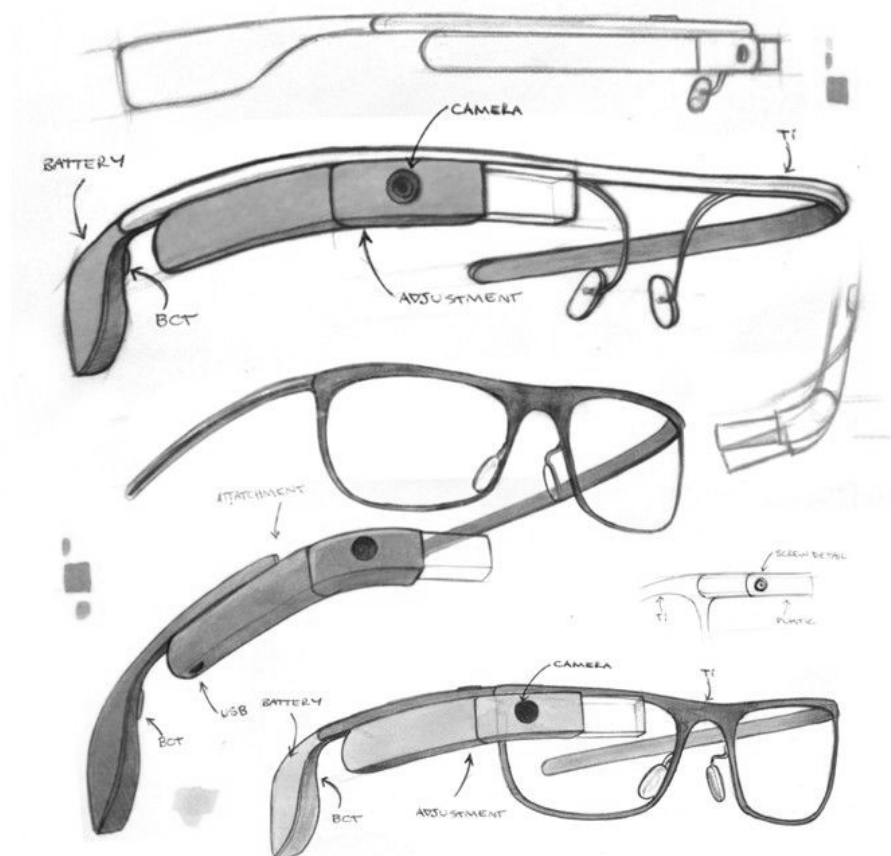
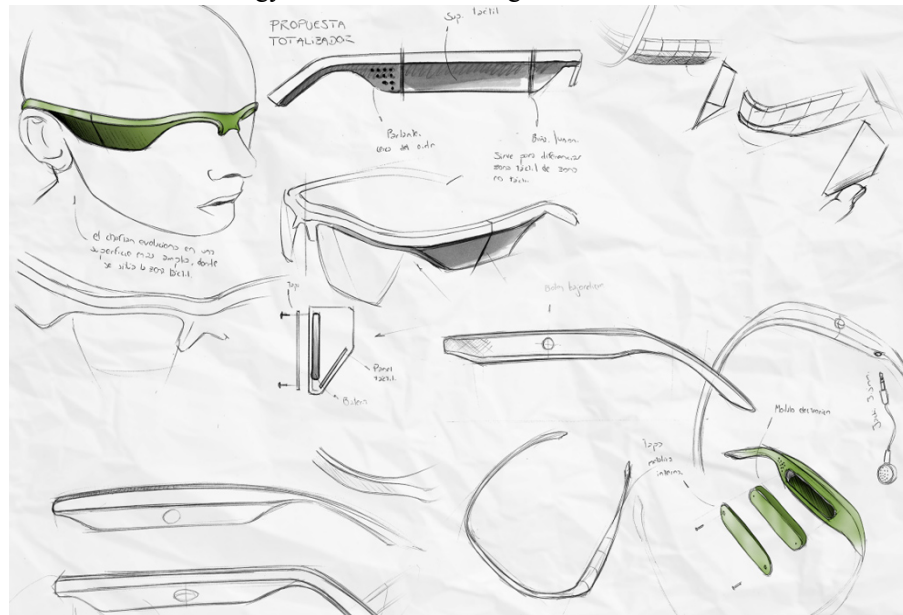
- 12 Show the designing of an iron box going through the various stages of the design process. Use hand sketches to illustrate the processes. (14)



Module 2

- 13 Illustrate the design thinking approach for designing a wearable technology for a college student. Describe each stage of the process. Illustrate the solution using sketches. (14)

The wearable technology considered is smart glass



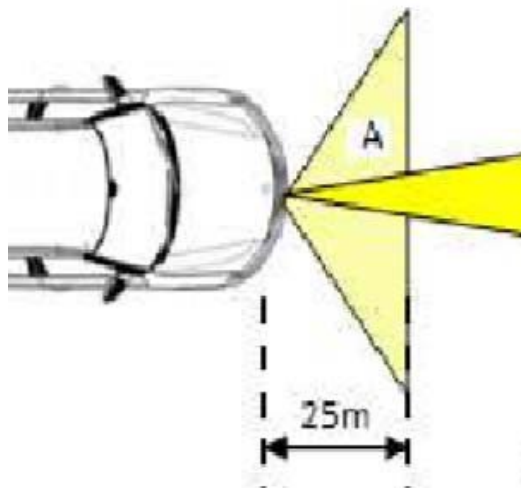
(14)

0800EST200122004

- 14 Some of the vehicle drivers do not dim the headlights when facing another vehicle at night. (14)
Empathize about this design problem and arrive at a solution using the design thinking process. Illustrate the solution using sketches.

Headlights of vehicles pose a great danger during night driving. The drivers of most vehicles use high, bright beam while driving at night. This causes a discomfort to the person travelling from the opposite direction. He experiences a sudden glare for a short period of time. This is caused due to the high intense headlight beam from the other vehicle coming towards him from the opposite direction. We are expected to dim the headlight to avoid this glare. This glare causes a temporary blindness to a person resulting in road accidents during the night. To avoid such incidents, we have fabricated a prototype of automatic headlight dimmer.

This automatically switches the high beam into low beam thus reducing the glare effect by sensing the approaching vehicle. It also eliminates the requirement of manual switching by the driver which is not done at all times.



a.) Both Vehicles at High Beam



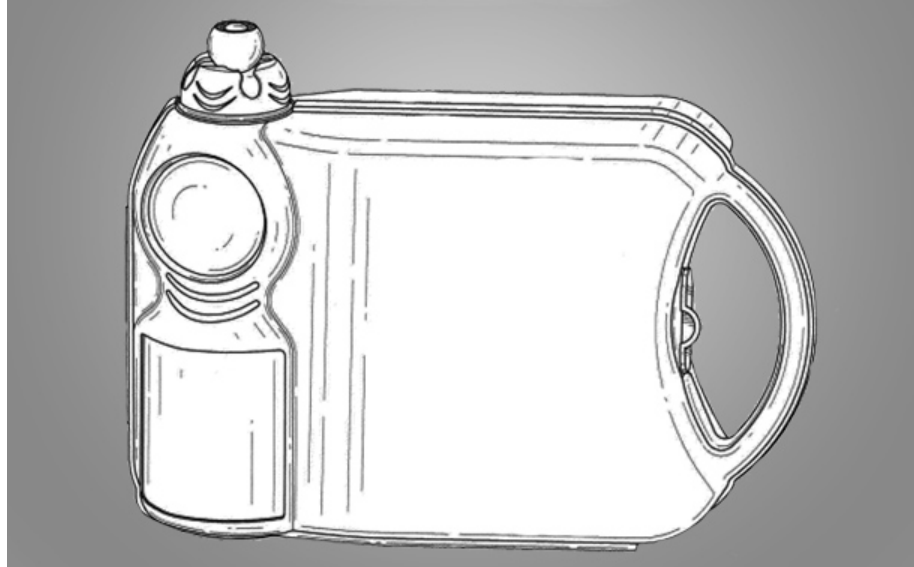
b.) Both the vehicles at low beam



c.) After passing both vehicles become high beam.

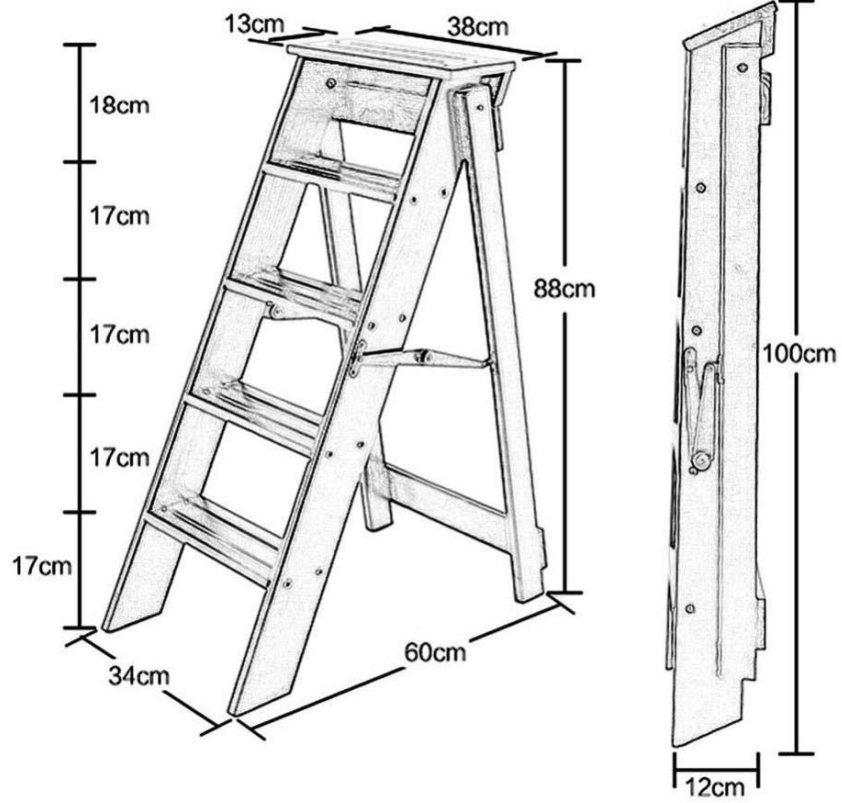
Module 3

- 15 Design an integrated water bottle with lunch box. Draw the detailed 2D drawings of the same with design detailing, material selection and dimensions. (14)
Use only hand sketches.



- 16 Prepare a technical report for a newly designed portable ladder with neat sketches for presenting to a client.

(14)



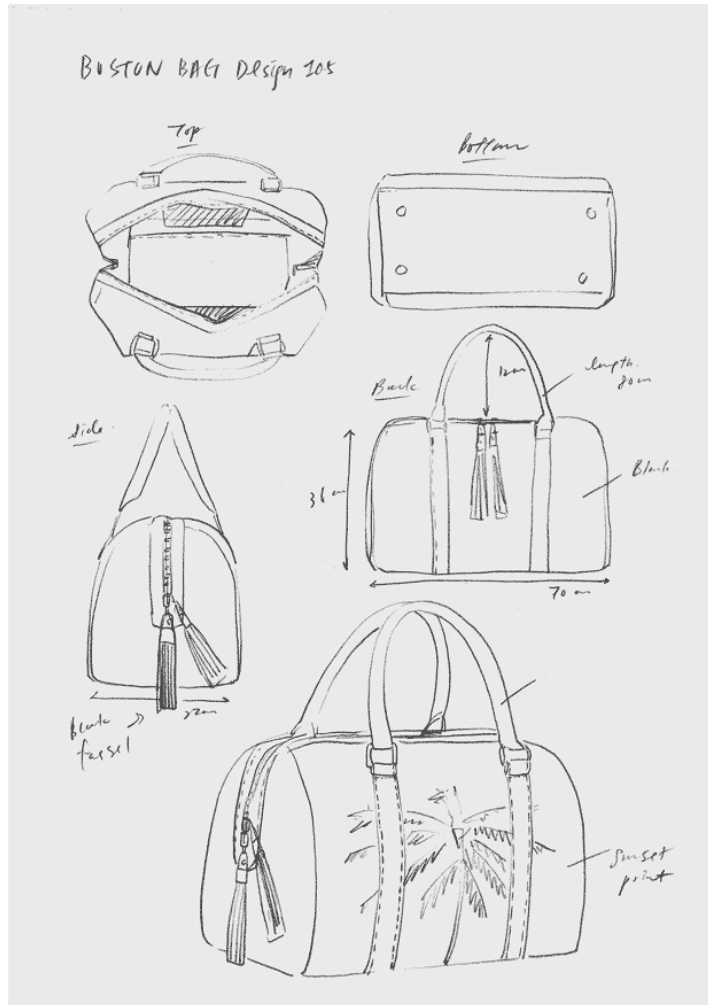


Module 4

- 17 Show the development of a nature-inspired design for a fashionable umbrella based on a banana leaf. Use hand sketches to support your arguments. (14)

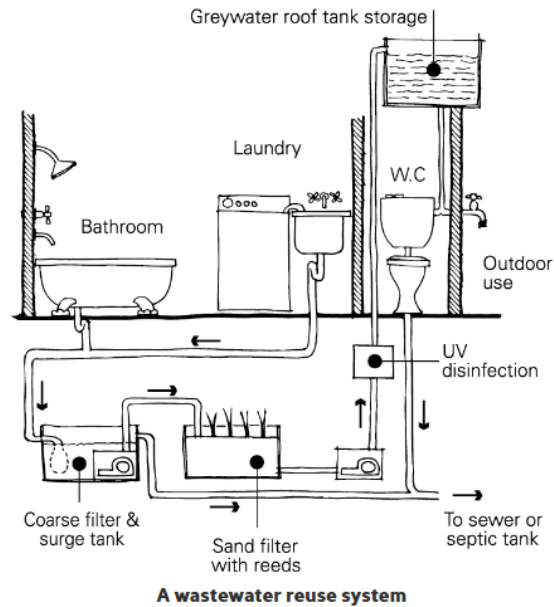


- 18 Develop some design modification for sports utility bag, to improve its functionalities as well as product value. Sketch the design. (14)



Module 5

- 19 Design a sustainable piping network for reuse of water in a residential building enabling water conservation. Sketch the design. (14)



- 20 Design a door handle with a lock which is easy to use. Use hand sketches and give rationalization for the various features in the design. (14)

